



TRIPLE POWER SUPPLY

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This low-cost, multipurpose power supply fulfils the requirements of almost all laboratory experiments. Nonetheless, it can be easily fabricated by hobbyists.

A single transformer is used to build this triple power supply. Regulator IC LM317 generates variable power supply of 1.25 to 20V, 1A. The dual $\pm 12V$, 1A power supply is generated by regulators 7812 and 7912. Similarly, dual $\pm 5V$, 1A power supply is generated by regulators 7805 and 7905. 'On'/'off' switches (S2 through S4) select the required power supply. Variable power supply is used to study the characteristics of devices. Fixed +5V power supply is used for all digital, microprocessor and microcontroller experiments. Dual $\pm 12V$ power supply

is used for op-amp-based analogue circuit experiments.

Fig. 1 shows the circuit of the triple power supply, while Fig. 2 shows the pin configuration of the regulators used in the circuit. Transformer X1 steps down the mains power to deliver the secondary output of 18V-0-18V. The transformer output is rectified by full-wave bridge rectifier BR1, filtered by capacitors C1, C2, C3, C7 and C8, and regulated by IC1 through IC5. Regulator IC1 (LM317) provides variable voltages (1.25 to 20V), while IC2 and IC4 provide regulated +12V and -12V, respectively. The output of IC2 is fed to regulator IC3 (7805), which provides fixed +5V. Similarly, the output of IC4 is fed to regulator IC5 (7905), which provides fixed -5V. Capacitors C4 through C6, and C9 through C11, are used for further filtering of ripples

in positive and negative regulated power supplies. LED1 glows to indicate that +5V is available, while LED2 indicates that -5V is available.

Switch S1 is used for mains 'on'/'off'. Using switches S2 through S4, any of the three supplies can be independently turned off when not required in a particular experiment. This reduces unnecessary power dissipation and increases the life and reliability of the power supply. Since the circuit uses three terminal regulators, only capacitors are required at the input and output. The use of few components makes the circuit very simple. The three terminal regulators have heat-sink provision to directly deliver 1A output current. To ensure the maximum output, do not forget to

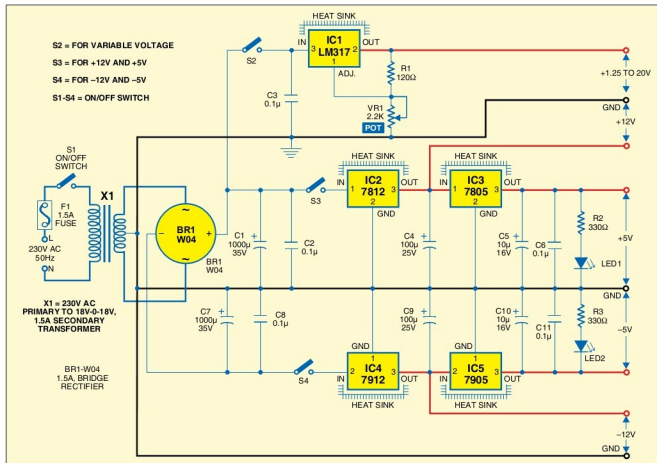


Fig. 1: Triple power supply